

When we say hardware (in terms of music), we mean the instruments. For music to flow, the instruments need to be in the right shape; they need to be tuned to perfection. Also, you must use the right instrument because if you want the sound of flute then a trumpet isn't going to help much.

Then we have the software, and this is analogous to the musician playing the instrument. It simply doesn't matter how great an instrument is if the player can't play it well. With a clueless or unskilled musician, all that comes out is plain noise.

And lastly, when we say synchronization, we mean the coherence or harmony among the players. For a tune to take shape, all musicians in the orchestra should know the notes they need to play and when to play each note so that they are in sync with the rest of the orchestra. Without this synchronization, you have wondrous cacophony.

Being a smart reader of Digit, you would probably have figured out what we mean by the analogy we just drew. If you are still clueless, here is what we mean to say: for a computing system to operate, there must exist the proper hardware in the right configuration, a well programmed software (firmware and drivers) that drives the hardware and applications should know how to operate on the hardware. Different components of the hardware and software must operate in sync with each other so that the system as a whole operates perfectly.

Given that analogy, the very goal of an SoC is to behave like an orchestra. An SoC contains several hardware components in a single package (or chip) so that they operate as coherently as possible. We are going to learn how that's done and in this chapter, we are going to focus on the hardware components that form an SoC.

The Jargon

Let's first clear up some terms and types of integrations that are popular today. We present them to you because the lines are blurred and there are multiple terms with relatively close meanings. You might come across these terms all too often while reading this book or when researching on your own.

1. **Chip:** Another name for an IC or Integrated Circuit.
2. **Integrated Circuits:** Miniaturised form of a larger circuit where the entire circuit is built on one die and placed inside a package.
3. **Die:** A die is a piece of semiconducting material (often called wafer or substrate) over which the circuit is etched.
4. **Substrate:** The film of semiconductor material on which a circuit is deposited. A Die is a base with a substrate layer on top of it, on which